

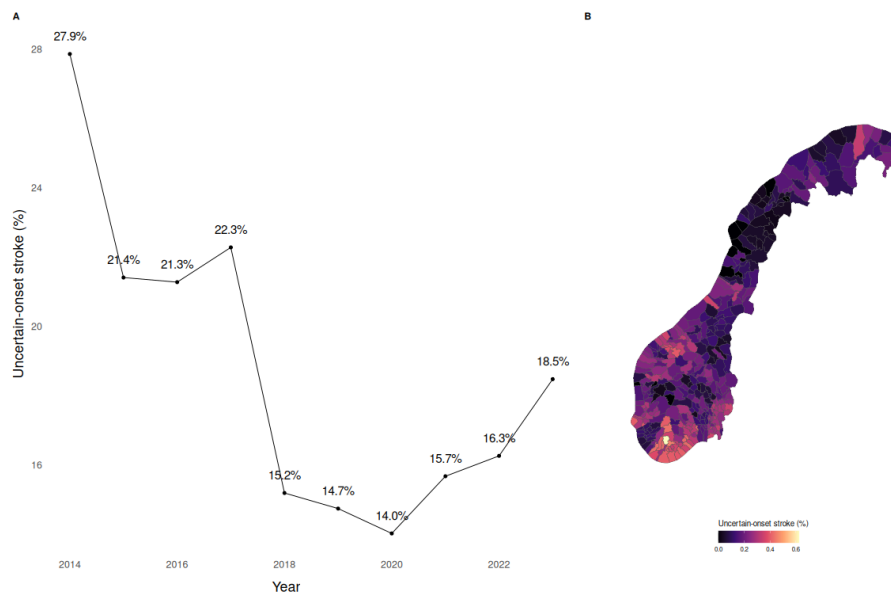
1 Results

The study population included 67,845 patients with stroke registered in the Norwegian Stroke Registry between 2014 and 2023, of whom 12,612 (18.6%) had uncertain-onset stroke (table 1). Most patient characteristics were similar between patients with uncertain-onset stroke and all other strokes, with standardized mean differences generally below 0.10. The largest differences were observed for living situation, and thrombolysis, while differences in demographic characteristics, stroke type, and cardiovascular history were modest. Living alone was more common among patients with uncertain-onset stroke (45.2% vs 37.2%), and they recieved thrombolysis less frequently (1.3% vs 22.0%).

Crude temporal and geographic variation in uncertain-onset stroke

The crude proportion of uncertain-onset stroke varied over time, with the heigest proportion in 2014 (27.9%) and the lowest in 2020 (14.0%). The proportion declined during the early years of the study period, with a marked reduciton between 2017 and 2018. It remained relatively low through 2020, followed by a gradual increase to 18.5% in 2023 (figure 1A).

Figure 1: Temporal and geographic variation in uncertain-onset stroke. A) Annual proportion of patients with uncertain-onset stroke, 2014–2023. B) Municipality-level geographic variation in uncertain-onset stroke.



In addition to the temporal variation, substantial geographic variation in uncertain-onset stroke was observed across municipalities, ranging from 0% to 62% with a median of 16.7% (IQR 10.2% and 25.8%) (Figure 1B). Areas with relatively high and low proportions appeared to cluster together geographically rather than an even distribution across the country.

Patient and provider factors and geographic variation

The base model only included year and municipality-level spatial effects. The municipality-level effects displayed a similar geographic pattern as the crude analysis, with the municipality-level estimates smoothed towards the mean (Figure 2A). The temporal pattern was also similar to the crude analysis, with the odds of uncertain-onset stroke being lower in all years compared to 2014 (supplementary table). The standard deviation of the municipality effect of teh base

Table 1: Characteristics of patients

	Uncertain onset (n = 12,612)	Other stroke (n = 55,233)	SMD
Age, mean (SD)	73.89 (13.4)	73.31 (13.4)	0.043
Female	5,945 (47.1)	25,337 (45.9)	0.025
Stroke type, Infarction	10,813 (85.7)	47,360 (85.7)	0.009
Previous TIA	880 (7.0)	4,356 (7.9)	0.043
Previous MI	1,482 (11.8)	6,816 (12.3)	0.020
Atrial fibrillation	2,917 (23.1)	13,119 (23.8)	0.015
Diabetes	2,438 (19.3)	9,234 (16.7)	0.069
Antihypertensive medication	6,508 (51.6)	28,464 (51.5)	0.040
Lipid-lowering medication	3,700 (29.3)	16,821 (30.5)	0.050
Thrombolysis	161 (1.3)	12,148 (22.0)	0.685
Smoking status			0.086
Never smoker	4,673 (37.1)	20,966 (38.0)	
Current smoker	2,717 (21.5)	10,234 (18.5)	
Former smoker	2,998 (23.8)	14,538 (26.3)	
Unknown	2,224 (17.6)	9,495 (17.2)	
Level of Consciousness (admission)			0.087
Alert	10,856 (86.1)	47,236 (85.5)	
Mildly Drowsy	800 (6.3)	4,283 (7.8)	
Markedly Drowsy	349 (2.8)	1,617 (2.9)	
Unresponsive	432 (3.4)	1,699 (3.1)	
Unknown	175 (1.4)	398 (0.7)	
Pre-stroke modified Rankin Scale			0.098
No symptoms	6,884 (54.6)	31,923 (57.8)	
No significant disability	1,927 (15.3)	8,805 (15.9)	
Slight disability	1,369 (10.9)	5,319 (9.6)	
Moderate disability	931 (7.4)	3,829 (6.9)	
Moderately severe disability	385 (3.1)	1,612 (2.9)	
Severe disability	54 (0.4)	227 (0.4)	
Unknown	1,062 (8.4)	3,518 (6.4)	
Living situation			0.173
Living with others	6,192 (49.1)	31,614 (57.2)	
Living alone	5,701 (45.2)	20,548 (37.2)	
Institution/nursing home	376 (3.0)	1,490 (2.7)	
Unknown	333 (2.8)	1,581 (2.8)	
Education			0.055
No education	60 (0.5)	203 (0.4)	
Compulsory education	4,150 (33.2)	17,149 (31.3)	
Upper-secondary education	5,868 (46.9)	25,743 (47.0)	
Higher education	2,419 (19.3)	11,606 (21.2)	
Unknown	15 (0.1)	60 (0.1)	
Household income			0.068
Low	3,325 (26.4)	13,437 (24.3)	
M. low	3,138 (24.9)	13,278 (24.0)	
M. high	3,009 (23.9)	13,470 (24.4)	
High	3,092 (24.5)	14,885 (26.9)	
Unknown	48 (0.4)	163 (0.3)	

SMD = standardized mean difference; SD = standard deviation; TIA = transient ischaemic attack.

model was 0.696 (95% CrI 0.614–0.781), and the spatial mixing parameter ϕ was 0.976 (95% CrI 0.969–0.982), indicating that most of the geographic variation was spatially structured (table 2).

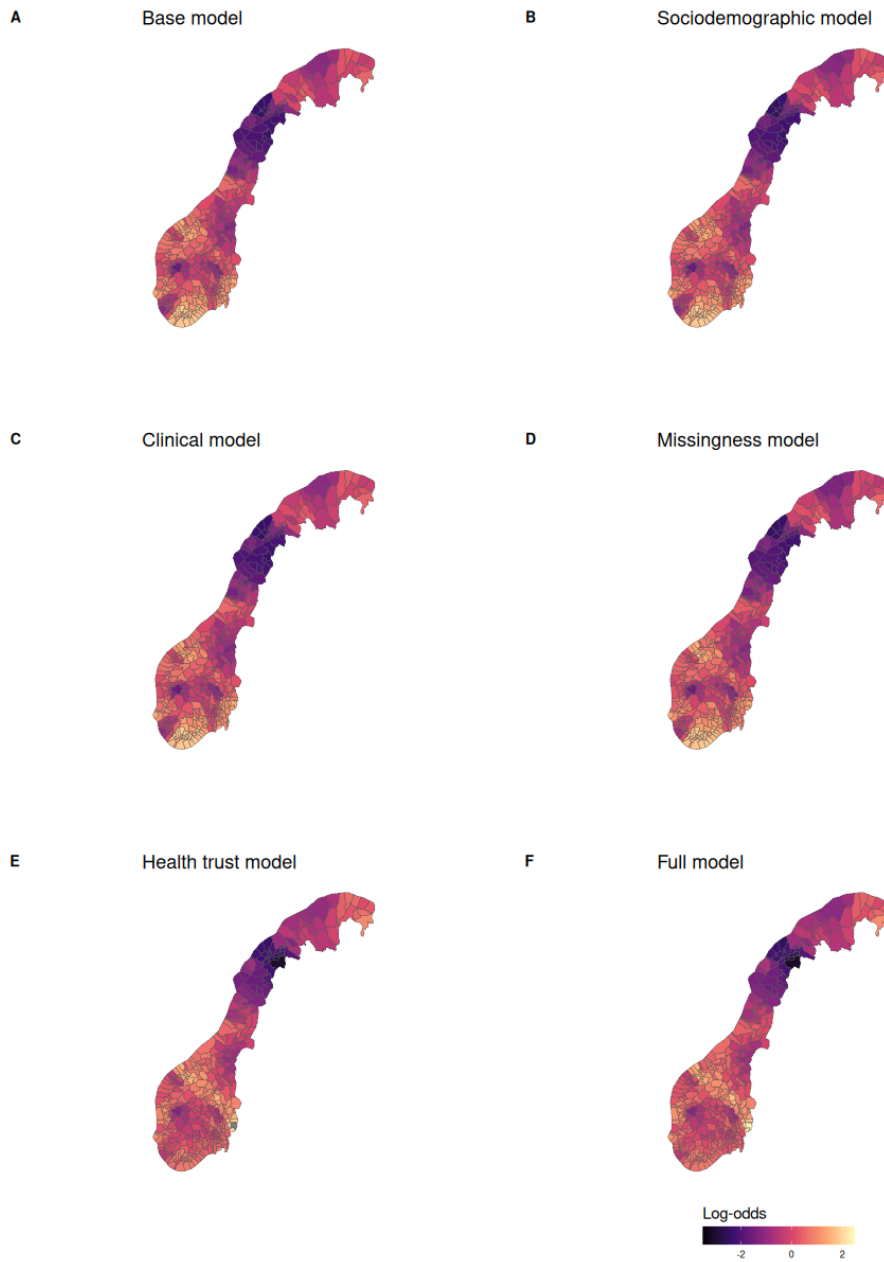


Figure 2: Geographic variation shown as the municipality-level estimates of uncertain-onset stroke as log-odds. A) Base model. B) Adjusted for sociodemographic characteristics. C) Adjusted for Clinical characteristics. D) Missingness of Wake-up status E) health trust. F) Full model.

The municipality level estimates were practically unchanged when adjusted for the patient-related sociodemographic and clinical characteristics. Compared to the base model, the municipality level log-odds estimates changed with median of 0.00 (IQR -0.02 - 0.02) for the sociodemographic model and -0.01 (IQR -0.03 - 0.03) for the clinical model (figure 2B-C). Adjustment for missing wake-up status produced a wider distribution of changes (median -0.01, IQR -0.08 - 0.08) (figure 2D). In contrast to the minimal visual changes seen after sociodemographic, clinical and missingness adjustment, the health trust (-0.08 median, IQR -0.48 - 0.48) and full models (-0.06 median, IQR -0.51 - 0.49) produced noticeable change in the spatial pattern (figure 2

E-F).

Changes in the municipality-level variation were reflected in the posterior standard deviation (SD) of the spatial effect. Compared to the base model (SD 0.696), the posterior standard deviation of the municipality effect of the sociodemographic (SD 0.697) and clinical models (SD 0.697) were unchanged. While higher spatial sd was observed for the missingness model (SD 0.775), lower spatial SD was obtained for the health trust model (SD 0.518) and full models (SD 0.612). The spatial mixing parameter remained high and largely unchanged across models, ranging from 0.969 to 0.976, indicating that the residual municipality-level variation remained predominantly spatially structured (table 2).

Table 2: Comparison of spatial models for uncertain-onset stroke

Model	Spatial SD	95% CrI	ϕ	DIC / WAIC
Unadjusted	0.696	0.614–0.781	0.976	61088.99 / 61089.08
Sociodemographic	0.697	0.616–0.783	0.976	60753.25 / 60752.12
Clinical	0.700	0.618–0.786	0.975	60780.37 / 60782.12
Missingness	0.775	0.683–0.871	0.969	54270.76 / 54269.53
Health trust	0.518	0.444–0.595	0.970	60710.35 / 60710.54
Full model	0.612	0.529–0.670	0.969	53638.95 / 53640.24

Spatial SD - the posterior standard deviation of the spatial effect. DIC = deviance information criterion; WAIC = Watanabe-Akaike information criterion. ϕ - the proportion of spatial variance attributed to the structured component.

The DIC and WAIC were lower for all models compared to the base model indicating improved model fit. The addition of patient-related factors improved the model fit without substantially altering the spatial pattern. Among the individual adjustment models, the largest reduction was observed for the missingness model (DIC 54270.76, WAIC 54269.53), accompanied with increase in the municipality level sd. Inclusion of health trust produced a smaller improvement in model fit (DIC, 60,710.35; WAIC, 60,710.54) but was accompanied by a marked reduction in spatial variation. The full model which included all patient and provider factors had the lowest DIC (53,638.95) and WAIC (53,640.24).

Geographic variation over time

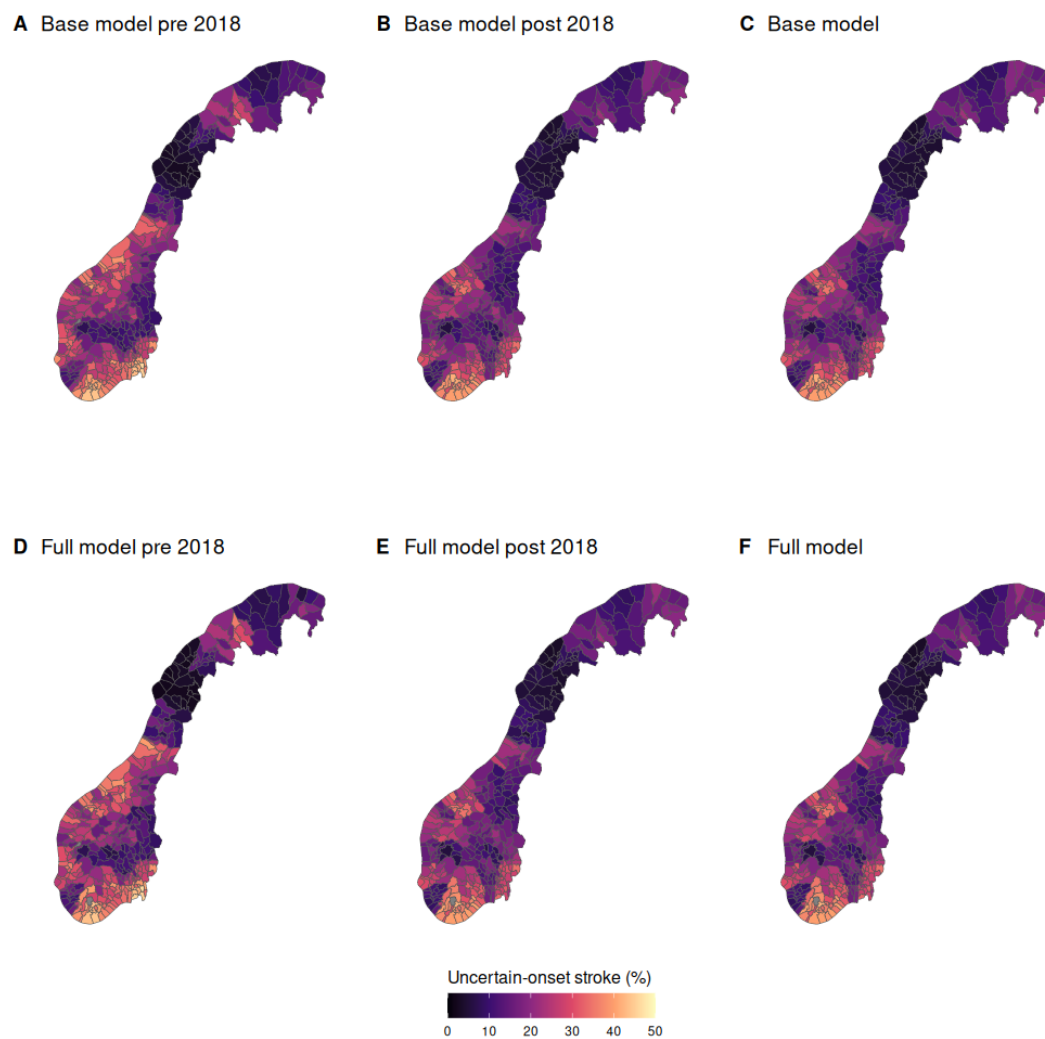


Figure 3: Geographic variation

Supplementary

Table 3: Factors associated with uncertain-onset stroke

	Predictor	OR (95% CrI)
Year	2015 vs 2014	0.67 (0.61–0.73)
	2016 vs 2014	0.64 (0.59–0.70)
	2017 vs 2014	0.63 (0.58–0.69)
	2018 vs 2014	0.37 (0.34–0.41)
	2019 vs 2014	0.36 (0.33–0.40)
	2020 vs 2014	0.24 (0.22–0.27)
	2021 vs 2014	0.28 (0.25–0.31)
	2022 vs 2014	0.30 (0.28–0.34)
	2023 vs 2014	0.41 (0.37–0.45)
Living situation	Living alone	1.19 (1.13–1.25)
	Institution/nursing home	0.80 (0.70–0.92)
Smoking	Current smoker	1.11 (1.04–1.18)
	Former smoker	0.95 (0.90–1.01)
Gender	Female	0.95 (0.90–0.99)
Income	Second quartile	0.98 (0.92–1.04)
	Third quartile	0.96 (0.89–1.02)
	Highest quartile	1.00 (0.93–1.07)
Education	Compulsory education	1.03 (0.85–1.26)
	Secondary education	1.04 (0.86–1.27)
	Higher education	1.07 (0.87–1.31)
Previous TIA	No	1.10 (1.01–1.20)
Previous MI	No	1.05 (0.98–1.13)
Atrial fibrillation	No	1.09 (1.03–1.15)
Diabetes	No	0.86 (0.81–0.91)
Antihypertensives	No	1.00 (0.96–1.05)
Lipid-lowering treatment	No	1.03 (0.98–1.09)
Pre-stroke mRS	mRS 1	1.03 (0.97–1.10)
	mRS 2	1.13 (1.05–1.23)
	mRS 3	1.11 (1.01–1.22)
	mRS 4	1.17 (1.02–1.35)
	mRS 5	1.41 (1.00–2.00)
Stroke type	Infarction	0.98 (0.91–1.04)
Consciousness	Mildly drowsy	0.72 (0.66–0.79)
	Markedly drowsy	0.78 (0.68–0.90)
	Unresponsive	0.89 (0.78–1.01)
wake-up-status Missingness	Individual	6.68 (6.36–7.02)
	Municipality-year	5.30 (4.29–6.56)
	Municipality-year (per 10pp)	1.18 (1.16–1.21)

TIA = transient ischaemic attack; MI = myocardial infarction; mRS = modified Rankin Scale. Bold estimates indicate 95% credible intervals excluding 1. The model additionally included municipality effects, age, and health trust. pp - percentage points.

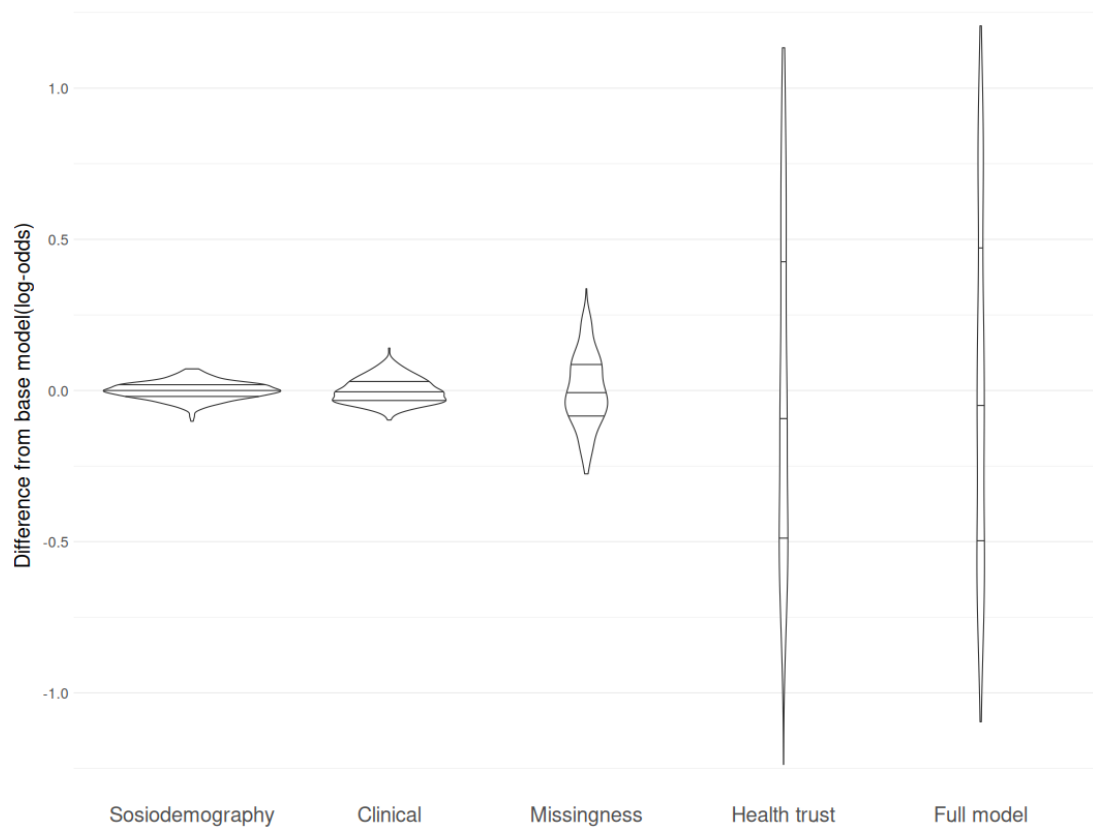


Figure 4: Violin plot of the mean difference in the proportion of uncertain-onset stroke between the base model and each of the different models.

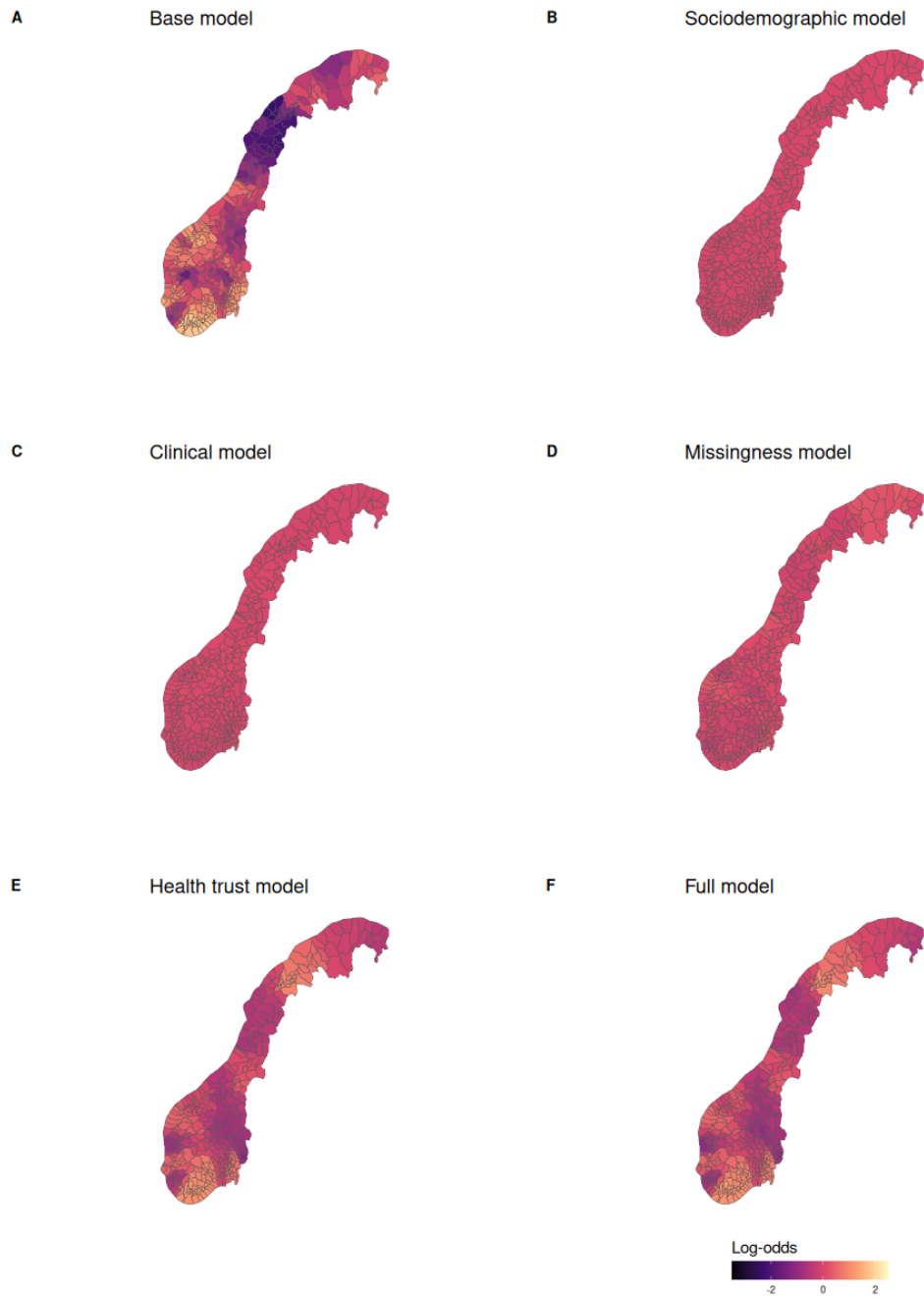


Figure 5: Geographic variation in the mean difference in the proportion of uncertain-onset stroke between the base model and each of the different models.

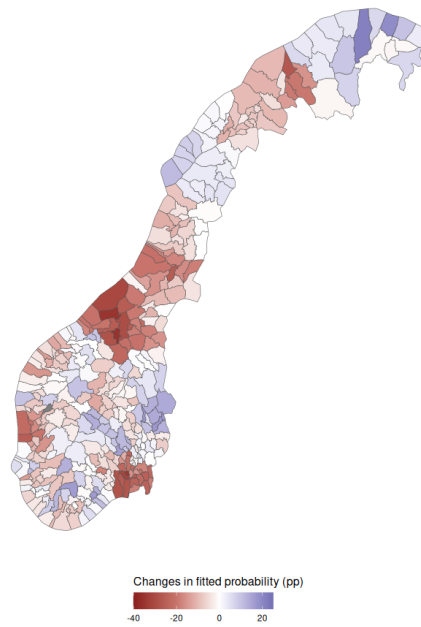


Figure 6: Geographic variation in the difference in the fitted proportion of uncertain-onset stroke between post treatment pathways for stroke minus the pre-treatment pathways